SECURE E-CERTIFICATE GENERATION & VERIFICATION SYSTEM

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Objective :

- To generate tamper-proof digital certificates for students using RSA-2048 digital signatures.
- To ensure certificate integrity using the SHA-256 hashing algorithm.
- To provide instant online verification of certificate authenticity using a unique Certificate ID.

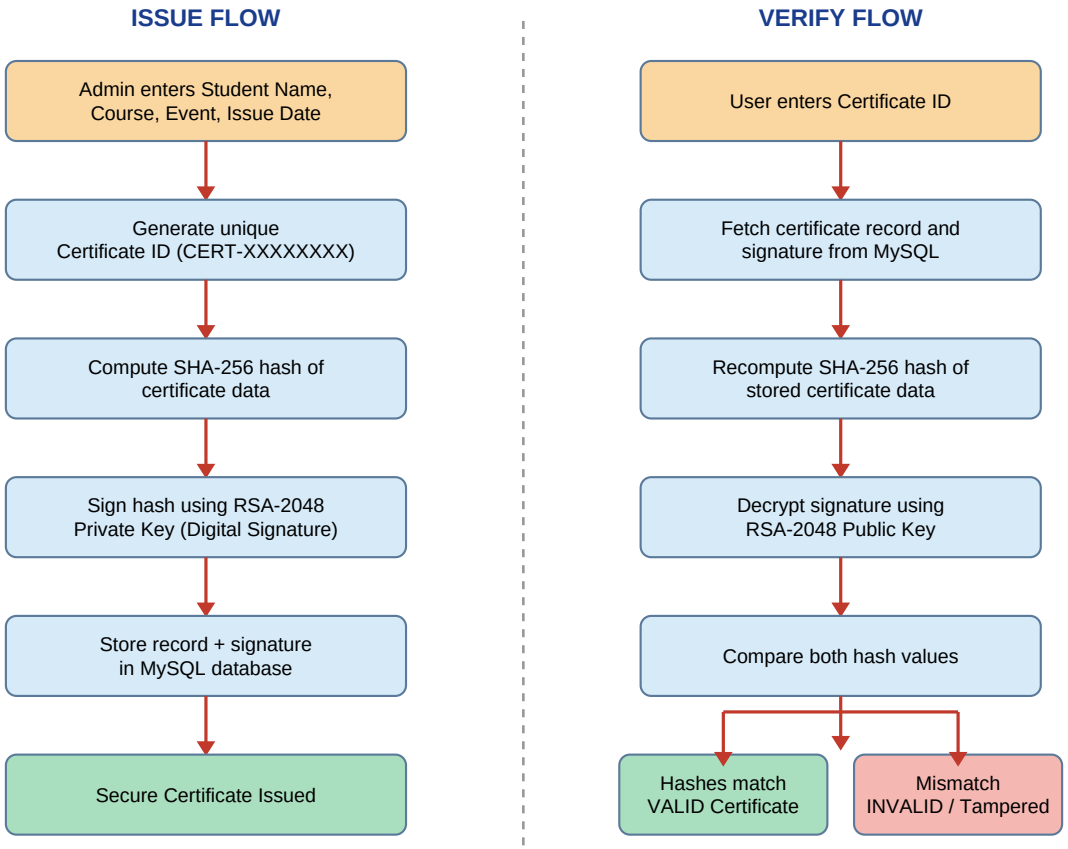
Abstract:

Paper certificates are easy to forge, slow to verify and difficult to manage. The Secure E-Certificate Generation & Verification System issues digital certificates that are protected by cryptography. When a certificate is issued, its details (student name, course/department, event name and issue date) are hashed with SHA-256 and the hash is digitally signed using an RSA-2048 private key. The certificate record and signature are stored in a MySQL database against a unique Certificate ID (CERT-XXXXXXXX). During verification, the system recomputes the hash and validates the signature with the public key; any modification to the certificate data makes the verification fail. This provides authenticity, integrity and non-repudiation for institutions and employers.

Innovation :

- Combines SHA-256 integrity checking with RSA-2048 digital signatures in a single, simple web application.
- Unique Certificate ID allows anyone to verify a certificate anytime, anywhere, without contacting the institution.

Methodology Diagram:



Working Principles:

- Hashing: the certificate data is converted into a fixed-length 256-bit digest using SHA-256; any change in the data produces a completely different digest.
- Digital signature: the digest is encrypted with the issuer's RSA-2048 private key, so only the issuer can create a valid signature.
- Verification: the stored signature is decrypted with the public key and compared with a freshly computed hash; a match proves the certificate is authentic and unaltered.
- Persistence: all certificate records are stored in a MySQL database and retrieved using the Certificate ID.

Screenshots:

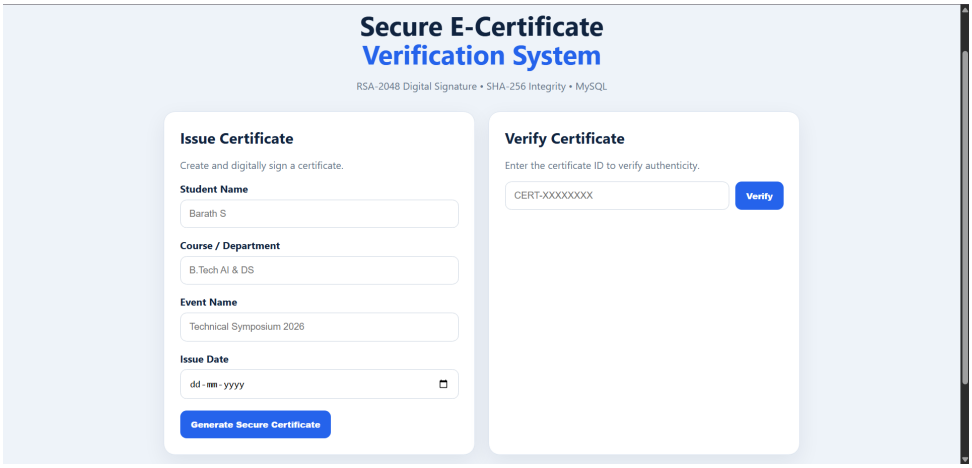


Fig: Issue Certificate and Verify Certificate interface

System Requirements:

H/W requirements:

- Hard Disk: 256 GB
- RAM: 4 GB
- Processor: Intel i3 or above
- Operating system: Windows

S/W requirements:

- Web browser (Chrome / Edge)
- MySQL Database
- RSA-2048 / SHA-256 libraries
- HTML, CSS, JavaScript

Outcome:

- The system produces digitally signed certificates that cannot be forged or altered, and reduces certificate verification from days to seconds.

Application Areas:

- Colleges and Universities
- Technical symposiums, workshops and online courses
- Companies (employee and background verification)

Challenges:

- Secure storage and protection of the RSA private key.
- Requires a proper network connection and a running database server.

Enhancements Identified:

QR code on each certificate for instant scan-and-verify, bulk certificate generation, e-mail delivery of certificates, and blockchain-based storage of certificate hashes.

Conclusion:

The Secure E-Certificate Generation & Verification System provides a reliable, fast and cryptographically secure way to issue and verify certificates. By combining SHA-256 hashing, RSA-2048 digital signatures and a MySQL database, it prevents certificate forgery and builds trust between institutions, students and employers.